

# Architecture A × processing unit P × lowering contract

A capability map, not a claim that every conceptual intersection has a production kernel.

| Flow organization A                                                                                                 | radix-2                                    | local radix-4                           | AIC / Cube DFT                          | specialized FFT core                    |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|
| <b>single-AIV, UB-resident FFT</b><br>Ts stage reuse + Ud/Td batch mapping<br>current production short path         | <b>PRODUCTION</b><br>planar stages         | <b>PRODUCTION</b><br>local stage fusion | <b>PROBE</b><br>not an FFT backend      | <b>FUTURE</b><br>requires callable core |
| <b>segmented long FFT</b><br>device/host boundary + consumer layout<br>current experimental path                    | <b>EXECUTABLE</b><br>segment core          | <b>EXECUTABLE</b><br>local fusion       | <b>NOT LOWERED</b><br>data path missing | <b>FUTURE</b><br>same flow contract     |
| <b>cross-AIV resident stage roles</b><br>producer/consumer overlap + handoff<br>method-space target, not production | <b>NOT LOWERED</b><br>needs real transport | <b>NOT LOWERED</b><br>same dependency   | <b>NOT LOWERED</b><br>AIC/AIV handoff   | <b>FUTURE</b><br>candidate only         |

## Lowering properties are evaluated for every intersection

AIV/AIC ownership · vector instruction shape · K-plane/planar layout · 32 B alignment · UB/workspace

MTE boundary · synchronization. “Vector” is a lowering mechanism, not a peer radix; whole-transform libraries remain external baselines.

illegal points never enter ranking